Amir R. Gabidulin

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EDUCATION

Washington State University, Vancouver, WA [*2021 – 2024*]
B.S. in Biology, Minor in Molecular Biology

Clark College, Vancouver, WA [*2019 – 2021*]
Associate of Science in Biology

RESEARCH EXPERIENCE

Post-Baccalaureate Researcher — Rudman Lab, WSU Vancouver [*May 2024 – Present*]

- Ran large-scale Drosophila phenotyping for the X-QTL/BSA experiment: 756,000 individuals across 125 replicate populations and 17 genetic and environmental contexts.
- Modified and integrated an X-QTL/BSA analysis pipeline (LSEI) for a repeated mapping experiment across genetic and environmental contexts; first-author manuscript in final co-author review.
- Led bioinformatic analysis for two host-microbiome response papers (Proc B 2025; Evolution, in revision): processing 16S rRNA data, inferential analysis of microbiome composition & association with phenotypes.
- Established and optimized pool-seq DNA extraction and quality-assessment protocols for large-scale genomic studies.

Undergraduate Researcher — Rudman Lab, WSU Vancouver [*Oct 2022 – May 2024*]

- Developed a machine-learning–based phenotyping pipeline (published as first author in G3).
- Contributed to the field mesocosm evolve-and-resequence experiments.
- Built and administered a local HPC for large-data storage and processing.

PUBLICATIONS

Peer-Reviewed

- **Gabidulin, A. R.†**, & Rudman, S. M. (2025). MLDAAPP: machine learning data acquisition for assessing population phenotypes. *G3: Genes, Genomes, Genetics*, 15(10). doi.org/10.1093/g3journal/jkaf173
- Shahmohamadloo, R. S., **Gabidulin, A. R.†**, Andrews, E. R., Fryxell, J. M., & Rudman, S. M. (2025). A test for microbiome-mediated rescue via host phenotypic plasticity in *Daphnia*. *Proceedings of the Royal Society B*, 292(2044). doi.org/10.1098/rspb.2025.0365
- Prileson, E. G., Campagnari, B., Clare, C. I., **Gabidulin, A. R.**, Shahmohamadloo, R. S., & Rudman, S. M. (2025). Overwintering drives rapid adaptation in *Drosophila* with potential costs to insecticide resistance. *Evolution*. doi.org/10.1093/evolut/qpaf205

Preprint / In-progress

- **Gabidulin, A.R.†,** Shahmohamadloo, R.S., Macdonald, S.J., & Rudman S.M. (2026) Repeated genotype-phenotype mapping reveals context dependency and the limits of genetic maps for predicting genomic adaptation. **Manuscript complete; in final co-author review, preprint forthcoming (09/2026)**
- Shahmohamadloo, R. S., **Gabidulin, A. R.†**, Andrews, E. R., & Rudman, S. M. (2025). Microbiome evolution plays a secondary role in host rapid adaptation. doi.org/10.1101/2025.06.27.661976. **Evolution - In revision (09/2026)**

† Led computational analyses.

TECHNICAL SKILLS

Programming: Python, R, Bash (proficient); C++, C#, Java (familiar).

Genomics & Statistics: haplotype imputation for pool-seq (LSEI & HAFpipe); pool-seq X-QTL/BSA mapping; 16S pipeline (QIIME2); variant-calling pipeline (FASTQ to VCF); population genetic data analysis; machine learning (PyTorch)

Computing: HPC (Slurm; Snakemake); built and administered a lab HPC.

Wet Lab: Drosophila melanogaster husbandry and large-scale phenotyping; DNA extraction and QA.

Languages: English and Russian (bilingual fluency).

AWARDS & HONORS

- WSU Undergraduate Academic Excellence Award for Outstanding Research (2024)
- NSF Rising Scientist Grant, The Allied Genetics Conference (2024)
- WSU Vancouver Undergraduate Research Showcase Winner (2023, 2024)
- WSU Vancouver Undergraduate Research Fellows Award (2023)

SELECTED PRESENTATIONS

- Poster, The Allied Genetics Conference (2024)
- Poster, WSU Vancouver Research Showcase (2023, 2024)
- Poster, EVO-WIBO (2023)
- Poster, M. J. Murdock Charitable Trust Conference (2023)

COMMUNITY ENGAGEMENT & OUTREACH

- Department of Biology research outreach at iTech and HeLa High Schools (2024).
- “Research as an Undergraduate” seminar, WSU Vancouver (2024).
- “Peak Your Interest” research podcast, WSU Vancouver (2023).